

Omni-Compass: five-reviewer consensus audit

Repository commit 1e0f4c3 (the delivered zip), audited 27 September 2026. The audit reran the zip from a clean copy. `verify.py` passed. The held-out result files regenerated byte-identical. A fresh fault sample was drawn. The live lever proof passed on real Kubernetes in GitHub Actions run 36270589830 (19 of 19 checks).

Independence. The five reviews below were written by the same assistant that built most of this code. They are an internal pre-review, organised the way an external panel would organise one. They are not an independent referee report. Every grade cites evidence a third party can rerun.

Grades: A (ready as is), B (sound, with named gaps), C (usable only with material caveats), D (not supported).

1. The five reviewers

Computational auditor: correctness and reproducibility (grade A-)

Findings

- `verify.py` passed on a clean copy.
- The C++ core ran a 100,000,000-decision soak at about 2.3 μ s per decision, with no failures and flat memory.
- Python/C++ parity checks pass.
- The closure-law held-out run (41 losing cells) and the whole-body site league reproduced exactly.
- Every tuned setting was frozen by SHA-256 before its held-out run (`tuning/*PREREGISTRATION*.json`).
- Every simulation lever defaults to off, and was checked to leave earlier results unchanged, trace hash for trace hash.

Deductions

- Held-out seed sets were spent round after round (700201 $\rightarrow$ 700301 $\rightarrow$ 700501 $\rightarrow$ 700601 $\rightarrow$ 700801 $\rightarrow$ 700901 $\rightarrow$ 701001). Each set is used once. But choosing *which rounds to report* is itself a researcher degree of freedom.
- The bootstrap loss rule uses 30 scenarios per cell. Small true differences can be missed.

Mechanism-of-action reviewer: does the mathematics do the work? (grade A-)

Findings

- The six-state core is stable. Its Jacobian modes are $\lambda = -7.9, -3.1, -0.72, -0.20 \pm 0.98i$ and -0.17 , giving time constants of 1.3 to 60 decisions.

- The operating point is $U = 0.976$, inside the healthy basin.
- Assimilation and the equations each carry about half of every state move (`tools/mechanism.py`).
- The closure law (`omnicompass/closure.py`) implements the manuscript directly:
 - forward projection (Section 5c); - $F = G_0 + G_c$ with no correction in the interior and an inward correction near $\partial\Omega$, i.e. $G \cdot n \leq 0$ (Chapter 20); - release only after the turning point (Chapters 29-30); - a dual-bath level/rate tracker (Chapter 31).
- Two further pieces are physical derivations, not tuned knobs:
 - the warm-reserve horizon at the break-even `BOOT_TICKS / park_frac = 24` ticks; - square-root and M/M/c staffing.
- Ablation shows each piece moves the gauges it should. With the closure law off, Omni-alone energy is 26% worse than Kubernetes. With it on, energy is 20-22% better.

Deductions

- The six-state core feeds the closure law only through its push term (the release gate). The machine decisions are made by the closure law.
- The core's own controller output is not applied during governance (by design, and stated).

Editor: claims discipline and clarity (grade B+)

Findings

- The report separates **measured / simulated / emulated** results.
- Every negative is printed: 41 losing cells for C against all platforms, 3 then 0 for B, and the failed live runs.
- The engine interpretation notice is present.
- Superseded numbers are labelled.

Deductions

- The report has grown section by section (6, 6A, 6B, 6C). A buyer-facing edition needs one clean structure.
- The patience round against Kubernetes and OpenShift is in `tuning/` but not yet in the report.
- Several terms from the manuscript (conveyance, basin) are used without a one-line engineering gloss.

Underwriter: what can be relied on, and what could go wrong (grade B-)

Findings

- Live evidence on real Kubernetes:

- the least-privilege identity is proven with `kubectl auth can-i` receipts; - the kill switch restores every lever, including from a fresh process; - one lever failing no longer blocks the others (this bug was found live and fixed); - a fail-safe hands control back to native after repeated failures.

- The live run in LIVE_ALLMUSCLE_6 (B on kind) measured 47% fewer nodes and 20-22% better p95/p99.

Deductions and risks

- The live evidence is kind on one CI machine, with power modelled rather than metered.
- There is no production or customer data.
- Traffic shift assumes replicated services and is simulated only.
- CPU-frequency and power caps are not proven on hardware.
- Cooling writes to a stand-in controller.
- In the public cloud, parked machines still bill. The energy saving from the warm reserve applies on owned hardware.
- The competitor emulations reproduce documented behaviour on a fluid plant. Their production binaries may differ.

Appraiser: value and market position (grade B)

Findings

- B (on top of any platform) is the strongest commercial asset.
- Held-out: 0 losing cells after the fixes, and 27 of 28 platform-workload pairs strictly better. - Four-cluster sites: 0 losses on all 7 platforms.
- C (Omni alone), against Kubernetes/OpenShift shops:
- energy 20-23% better; - machine-hours 34-38% better; - slowest responses 46-60% faster; - fewer pages under faults (14-25%).
- C at four-cluster sites: zero losses against CAST AI, Karpenter, Spot and GKE.
- The category (the capacity and cost-optimisation layer: HPA, VPA, Cluster Autoscaler, Karpenter, KEDA, CAST AI, Spot, Turbonomic) is a paid, established market.

Deductions

- Against the tight packers on single clusters, Omni alone roughly ties on energy. The measured advantage there is latency and stability, not the bill.
- Price-aware features (spot instances, instance types) are absent.
- There is no customer reference yet.

2. Grades by level of the code

Level	Path	Grade	Basis
Six-state core (Python + C++)	omnicompass/core.py, cpp/	A	stable modes, parity, 100M-decision soak, SHA-locked
Closure law and levers	omnicompass/closure.py, fleet/sim_slo.py levers	A-	manuscript laws in code, physical break-evens, defaults reproduce frozen results; per-workload settings are tuned
Governor, shield, fail-safe	omnicompass/adaptor.py, shield.py, omni_controller/controller.py	A-	invariants enforced, fail-safe tested with injected API outages
Live controller and levers	omni_controller/muscles.py, scripts/kind_levers.sh	B+	19/19 live checks, RBAC receipts, isolated kill switch; small cluster, three failed runs before pass (recorded)
Simulation plant and opponents	fleet/, hardware/schedutil.py	B	M/M/c response, boot delay, power and heat models, documented vendor emulations; fluid packing, no fragmentation
Benchmark method	tuning/, tools/protocol_bench.py	A-	development/held-out split, SHA pre-registration, paired bootstrap, fault protocol; many held-out rounds
Results against claims	tuning/*HELDOUT*.json, results/	B	B strong (0 losses); C 41/364 against all platforms at once, 10-30 against single incumbents
Reproducibility and packaging	verify.py, zip	A	clean-copy rerun identical
Documentation and report	docs/	B+	complete and honest; needs a single buyer edition
Commercial readiness	n/a	C+	no customer pilot, no metered power, no production run

Consensus: B+ overall.

- Engineering and mathematics: A-.
- Evidence strength: B. Simulation plus a small live cluster.
- Bankability: C+ until a customer pilot.

The single highest-value next step is a shadow-mode pilot on a real customer cluster with metered power.

3. Rerun this audit

```
pip install -r requirements.txt
python verify.py
python tools/mechanism.py
python tuning/closure_heldout.py
python tuning/site_league.py --heldout
python tools/protocol_bench.py 20
```

- tuning/closure_heldout.py must match tuning/CLOSURE_HELDOUT.json.
- tuning/site_league.py --heldout must match tuning/SITE_LEAGUE.json.
- Live: push a commit whose message contains [levers] (workflow live-kind-levers).